

Binding & Beyond Docking: Molecular Study on EGFR Mutations in Lung Cancer

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B. TECH IN BIOINFORMATICS

3rd year

(BATCH 2023-2027)

ABSTRACT

Non-Small Cell Lung Cancer (NSCLC) is one of the leading causes of cancer-related deaths worldwide, with mutations in the Epidermal Growth Factor Receptor (EGFR) playing a major role in uncontrolled cell proliferation and tumor progression. Gefitinib, a tyrosine kinase inhibitor (TKI), specifically targets aberrant EGFR activity; however, understanding how mutations alter its binding behavior and structural stability remains crucial for improving therapeutic designs. This project, **“Binding and Beyond: Docking to Dynamics (Molecular Study on EGFR Mutations in Lung Cancer)”**, computationally investigates the interaction of **Gefitinib** with both **wild-type and mutant EGFR receptors** through **molecular docking** and **molecular visualization** approaches. The workflow was divided into two modules:

Module 1: Molecular Docking — Wild-type and mutant EGFR structures were retrieved from the Protein Data Bank (PDB), and Gefitinib’s 3D structure was obtained from PubChem. Docking was performed using **PyRx (AutoDock Engine)** to predict binding affinities and poses, followed by interaction analysis using the **PyMOL**.

Module 2: Molecular Visualization — The best docked complexes were visualized using **PyMOL** to assess conformational stability under physiological conditions. to assess their **conformational orientation, active-site interactions, and binding pocket architecture**. Visualization enabled detailed analysis of **hydrogen bonds, hydrophobic interactions, and residue contacts** between **Gefitinib and the EGFR receptor**, providing structural insights into the **binding mechanism and stability** of the complexes.

The results demonstrated that **Gefitinib binds more strongly and specifically to mutant EGFR** than to the wild-type receptor, reflecting its therapeutic selectivity. Docking scores indicated higher binding affinity for the mutant complex, supported by additional stabilizing interactions at the mutation site. The Molecular Visualization further completed and confirmed these findings by clearly depicting **enhanced residue interactions and a better fit of Gefitinib within the ATP-binding pocket** of the mutant EGFR. The integrated docking-to-visualization approach thus provides a valuable framework for understanding mutation-driven drug selectivity and receptor stability, supporting future research in structure-based drug design and precision oncology.

Key Words: *EGFR, Gefitinib, Non-Small Cell Lung Cancer (NSCLC), Molecular Docking, Molecular Visualization, PyRx, AutoDock, , Binding, Structural Stability & Drug Selectivity.*

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CHAPTER 1- OBJECTIVES OF STUDY

1.1 OBJECTIVE OF PROJECT

To computationally investigate the interaction of Gefitinib (ligand/drug) with both wild-type and mutant EGFR receptors associated with non-small cell lung cancer (NSCLC) using molecular docking via, PyRx (uses AutoDock as the backend docking engine & inturn AutoDock utilizes Genetic Algorithm (GA)) and molecular visualization via, PyMOL, in order to visualize and understand how mutations alter receptor conformation, drug binding affinity, and structural stability. Thus, the study aims to provide molecular-level insights into the structural basis of EGFR–Gefitinib interaction, supporting the understanding of how EGFR mutations contribute to lung cancer progression and therapeutic response.

Key Outcome- Compared the interaction of Gefitinib with both wild-type and mutant EGFR to understand mutation-driven changes in binding and receptor behavior.

1.1.2 OBJECTIVE OF MOLECULAR DOCKING

To predict and compare the binding affinities, active-site interactions, and conformational differences between Gefitinib docked with wild-type and mutant EGFR receptors using PyRx software, which employs the AutoDock algorithm. Although Gefitinib primarily targets mutant EGFR receptors in NSCLC, molecular docking with the wild-type EGFR was also performed to establish a baseline reference for comparison. This enables assessment of how specific mutations alter the receptor's binding site architecture, drug affinity, and interaction patterns.

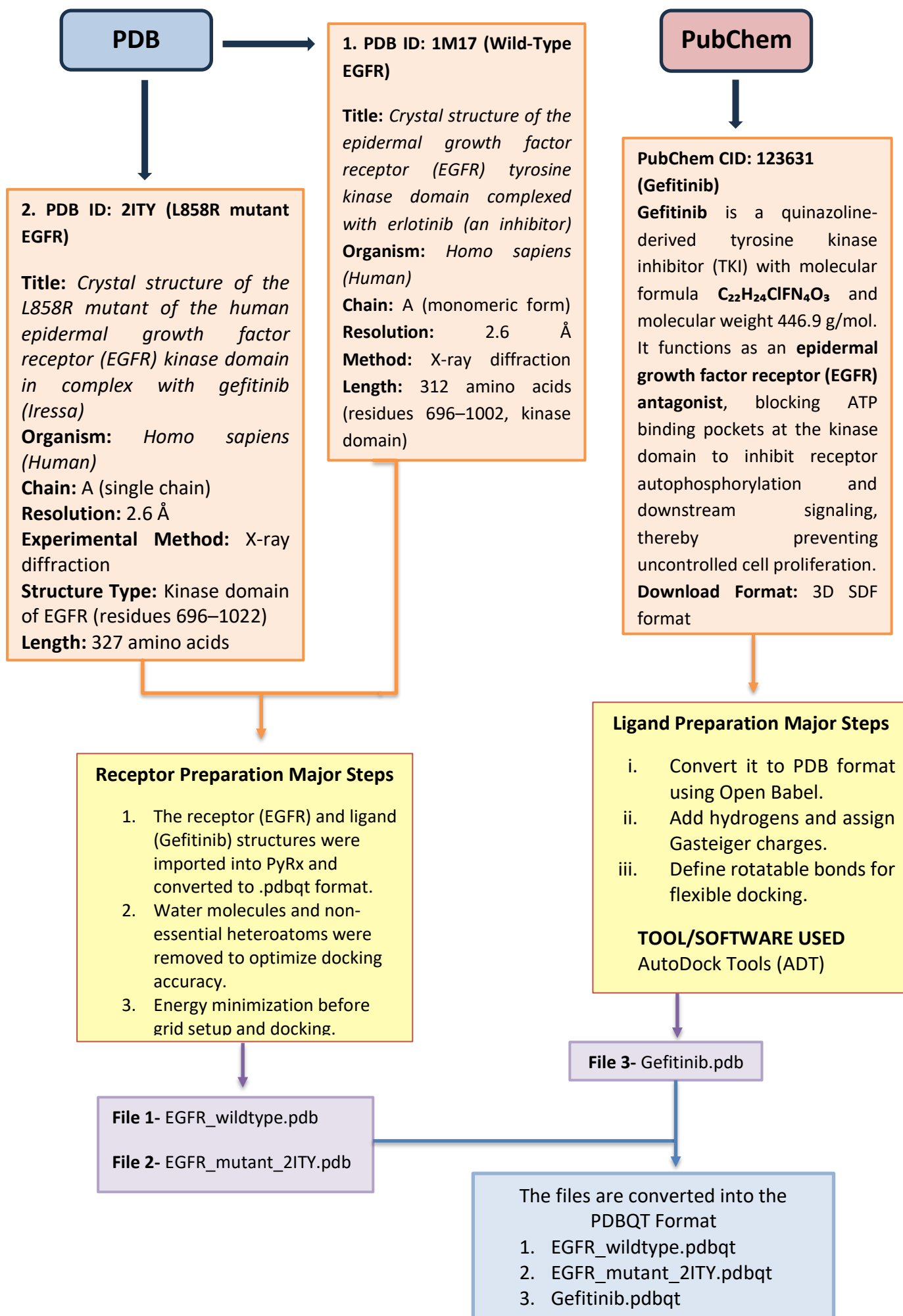
Key Outcome- Docking revealed stronger and more specific binding of Gefitinib with mutant EGFR compared to wild-type, highlighting altered active-site interactions.

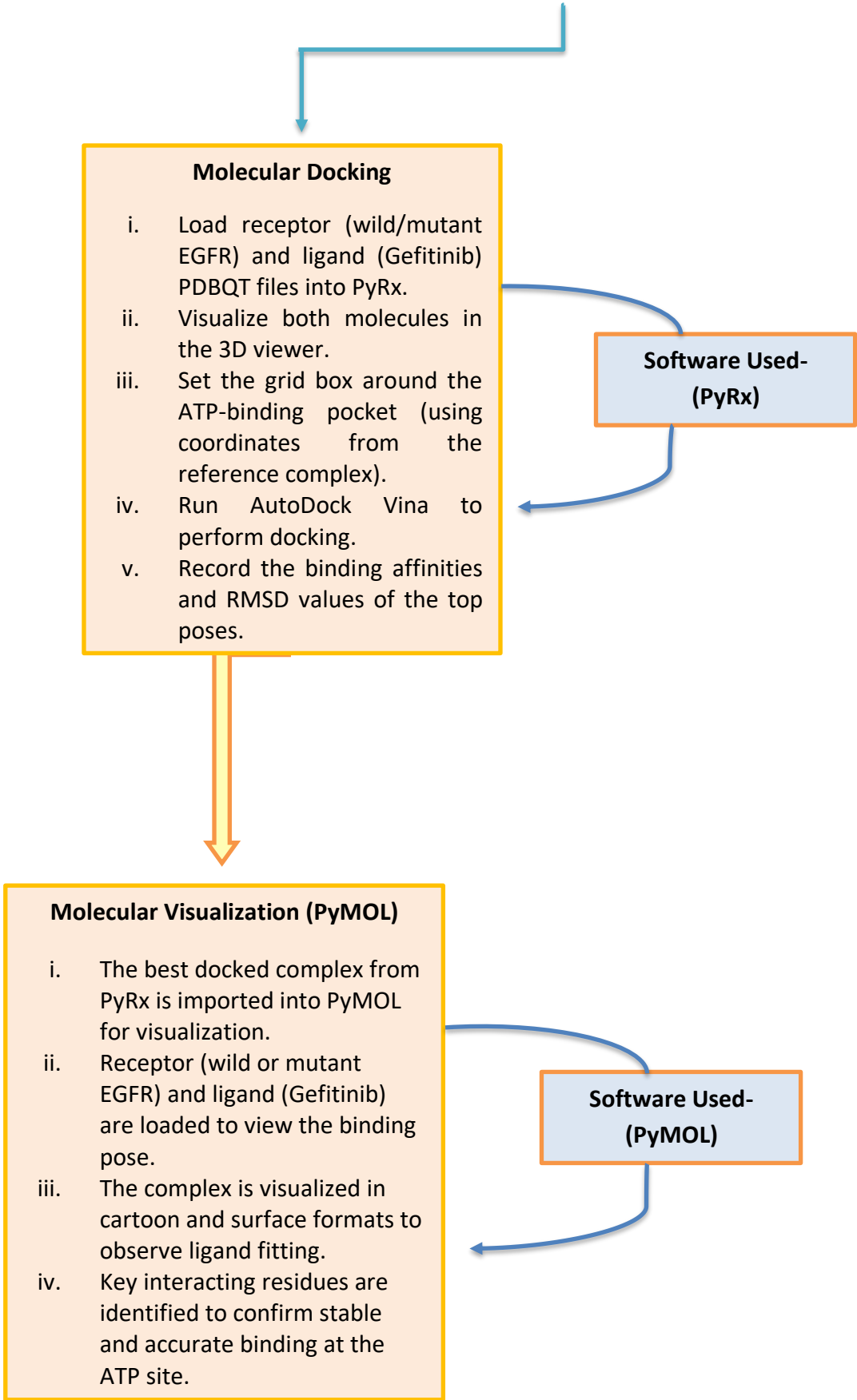
1.1.3 OBJECTIVE OF MOLECULAR VISUALIZATION USING PyMOL

To visualize and analyze the binding conformation and intermolecular interactions of the docked EGFR–Gefitinib complexes (both wild-type and mutant) using PyMOL. This module aims to provide a detailed structural insight into the ATP-binding pocket, identifying key residues involved in ligand binding and confirming the accuracy of docking results through 3D visualization. The visualization also facilitates interpretation of ligand fitting, interaction geometry, and binding pocket accessibility in different conformations of EGFR.

Key Outcome: PyMOL visualization confirmed that Gefitinib binds precisely within the ATP-binding pocket, interacting with critical residues responsible for its inhibitory action, with noticeable differences between wild-type and mutant EGFR binding conformations.

1.2. WORK FLOW OF THE PROJECT





```
graph TD; A[ ] --> B[Molecular Docking]; B --> C[Software Used- PyRx]; C --> B; B --> D[Molecular Visualization PyMOL]; D --> E[Software Used- PyMOL]; E --> D;
```

Molecular Docking

- i. Load receptor (wild/mutant EGFR) and ligand (Gefitinib) PDBQT files into PyRx.
- ii. Visualize both molecules in the 3D viewer.
- iii. Set the grid box around the ATP-binding pocket (using coordinates from the reference complex).
- iv. Run AutoDock Vina to perform docking.
- v. Record the binding affinities and RMSD values of the top poses.

**Software Used-
(PyRx)**

Molecular Visualization (PyMOL)

- i. The best docked complex from PyRx is imported into PyMOL for visualization.
- ii. Receptor (wild or mutant EGFR) and ligand (Gefitinib) are loaded to view the binding pose.
- iii. The complex is visualized in cartoon and surface formats to observe ligand fitting.
- iv. Key interacting residues are identified to confirm stable and accurate binding at the ATP site.

**Software Used-
(PyMOL)**

CHAPTER 2- METHODOLOGY

2.1 Retrieval of Receptor and Ligand Structures

The molecular docking and simulation study was performed using both the wild-type and mutant forms of the Epidermal Growth Factor Receptor (EGFR) as receptor molecules, and Gefitinib as the ligand (drug molecule).

2.1.2 Receptor Retrieval:

The three-dimensional crystal structures of the EGFR receptor proteins were obtained from the Protein Data Bank (PDB) — an open-access repository for experimentally determined protein and nucleic acid structures. The PDB provides structural data derived from X-ray crystallography and NMR spectroscopy in .pdb format, which can be further used for computational modeling and docking studies.

- **Wild-Type EGFR:** PDB ID – *1M17* (EGFR kinase domain complexed with erlotinib).
- **Mutant EGFR:** PDB ID – *2ITY* (EGFR kinase domain with L858R mutation complexed with Gefitinib).
- **Database Source:** <https://www.rcsb.org/>

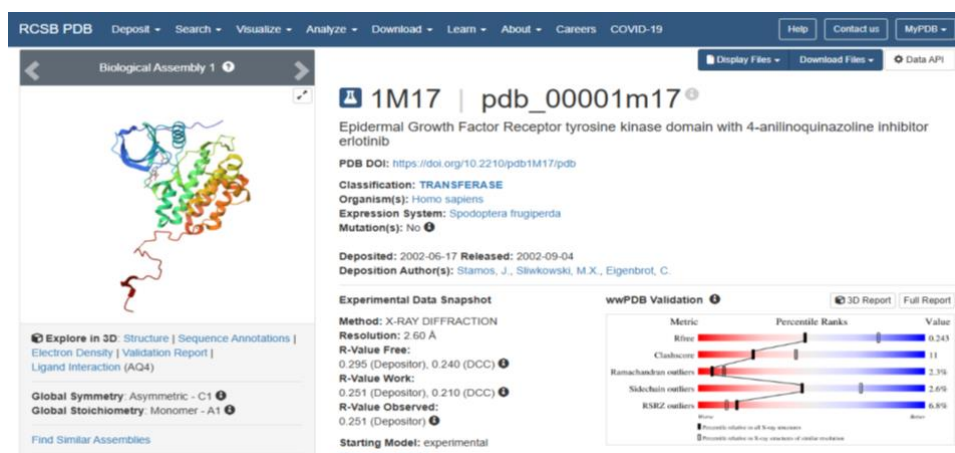


Figure:1 (wild-type receptor)

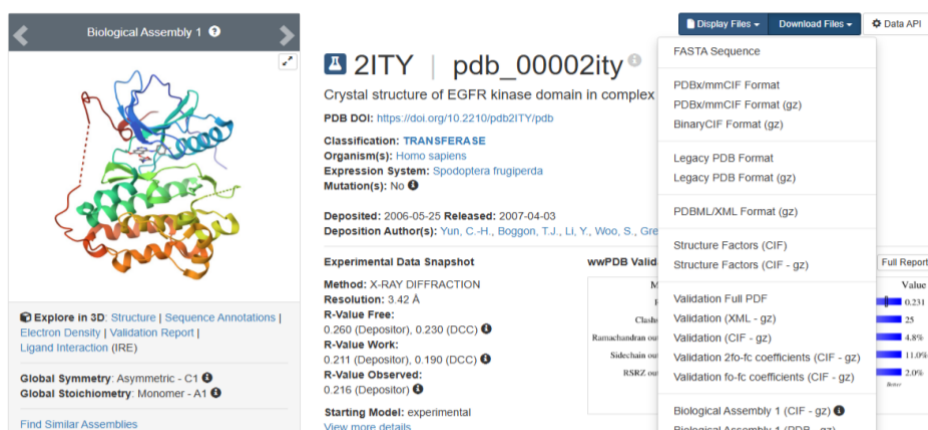


Figure:2 (mutant receptor)

2.1.3 Ligand Retrieval:

The **ligand molecule, Gefitinib**, a known tyrosine kinase inhibitor targeting EGFR, was retrieved from the **PubChem** database in .sdf format. PubChem is an official repository maintained by the National Center for Biotechnology Information (NCBI) that provides chemical structure, bioactivity, and molecular property data.

- **Ligand Name:** Gefitinib
- **PubChem Compound ID (CID):** *123631*
- **Database Source:** <https://pubchem.ncbi.nlm.nih.gov/>

The downloaded ligand file was then converted into a suitable format (.pdb) for further studies.

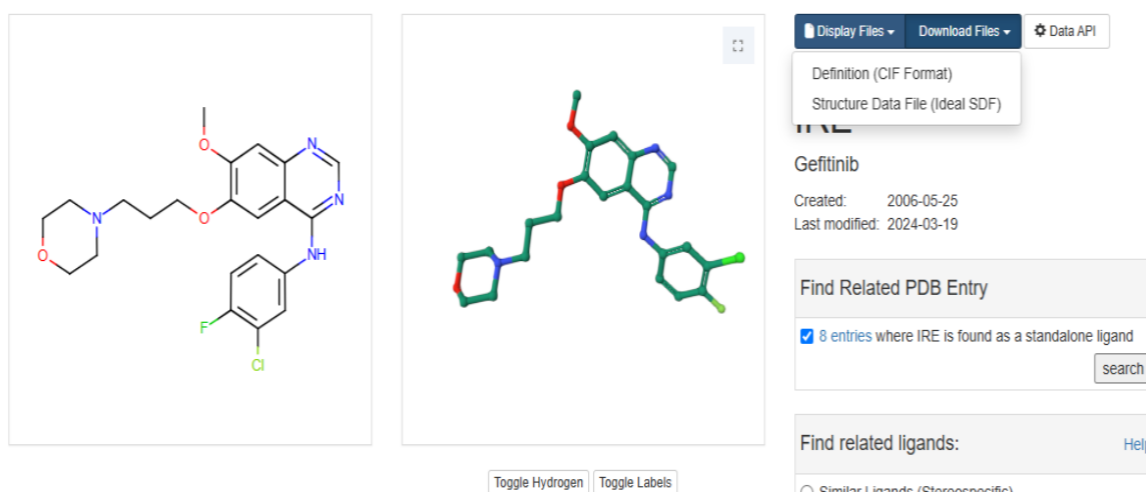


Figure: 3 (Gefitinib in PubChem)

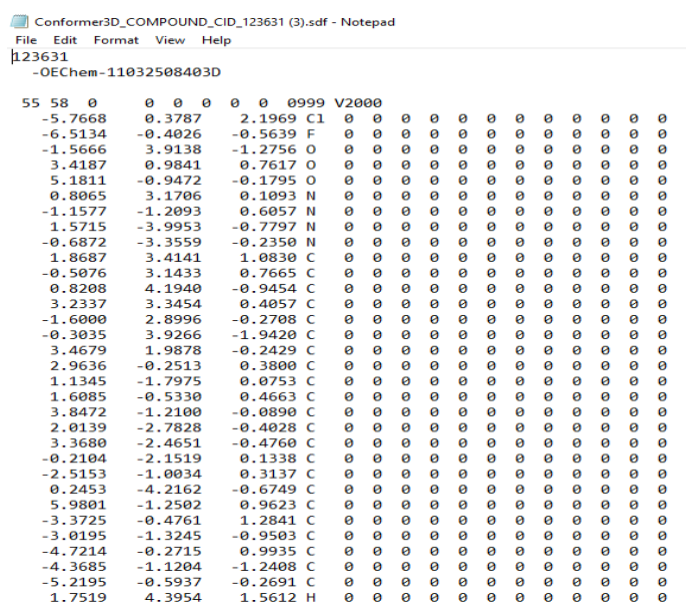


Figure: 4 (Gefitinib in .sdf file format)

Since, AutoDock Tools does not directly support .SDF files, the structure was converted into **PDB format** using **Open Babel**:

- Input: Gefitinib.sdf
- Output: Gefitinib.pdb

This conversion preserves the 3D conformation and bonding pattern of the ligand for further processing in ADT.

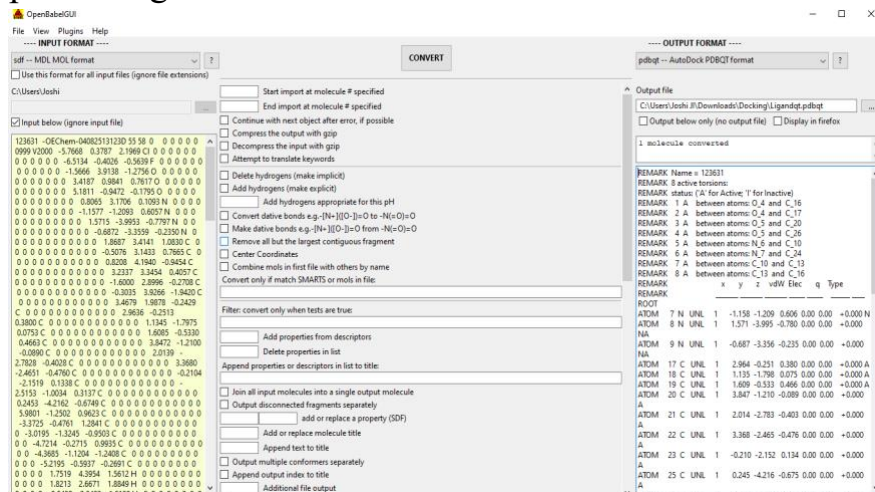


Figure-5: (Open Babel)

3.2 Receptor Preparation

The downloaded structures were preprocessed via, **ADT** (software) by removing water molecules, heteroatoms, and converted to the “**Apo Form**” means the receptor structure **without any bound ligand** (is used for docking) therefore, any co-crystallized ligands is removed before the further analysis.

During **receptor preparation**, the followed are removed:

- Water molecules (to eliminate noise and unnecessary interactions)
- Co-crystallized ligands (to free the binding site for new docking)

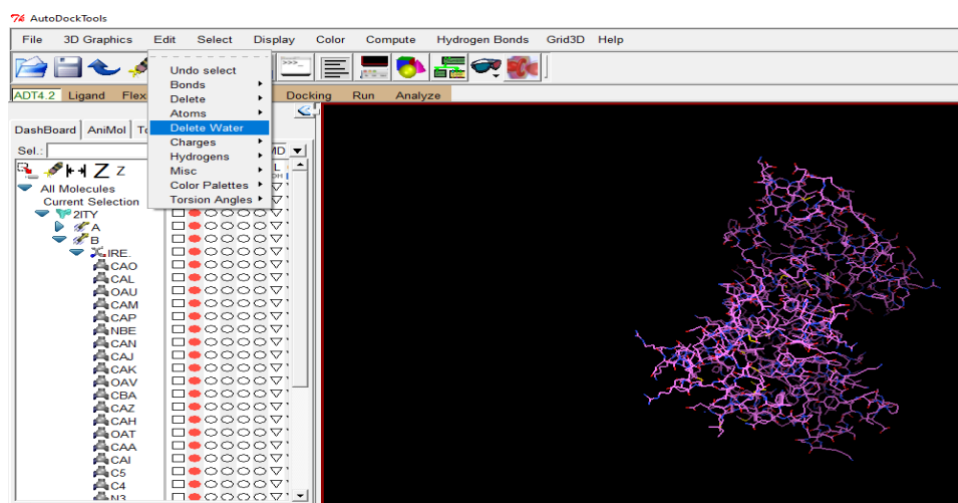


Figure- 6: (Water molecules are removed)

- Any extra heteroatoms not part of the protein (chain B (Gefitinib) not required)
- Added polar hydrogen atoms were added to the receptor to define hydrogen-bonding interactions and correct charge distribution for accurate docking results.
- Kollman charges were assigned to the receptor to accurately represent the electrostatic potential and ensure realistic ligand–receptor interaction modeling during docking. And finally save the file in PDBQT format.

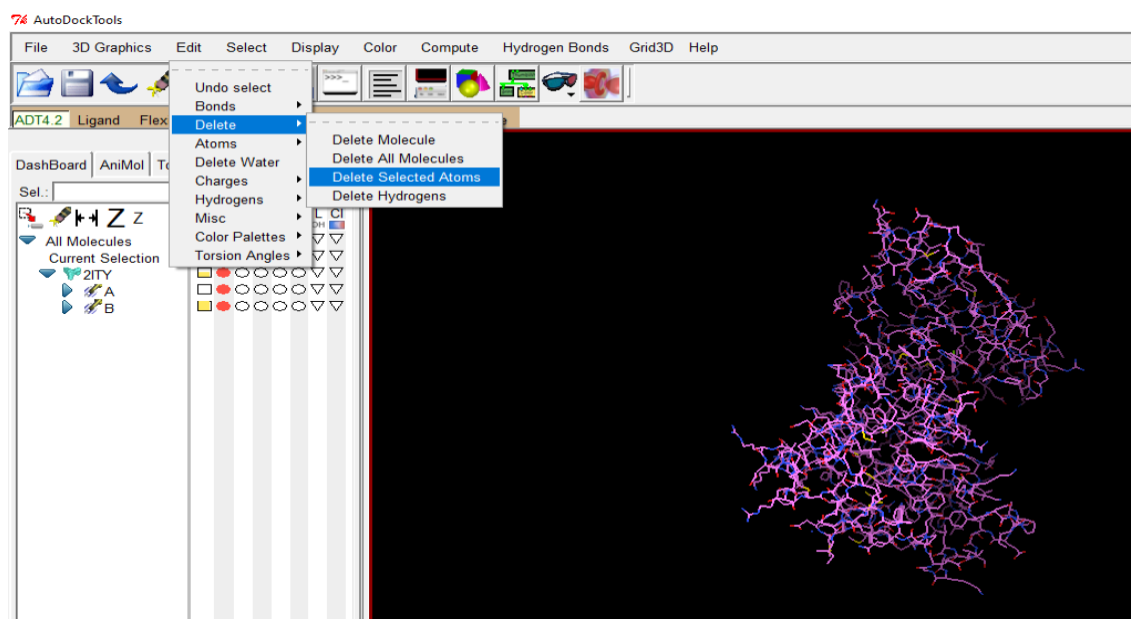


Figure- 7: (Heteroatoms are removed)

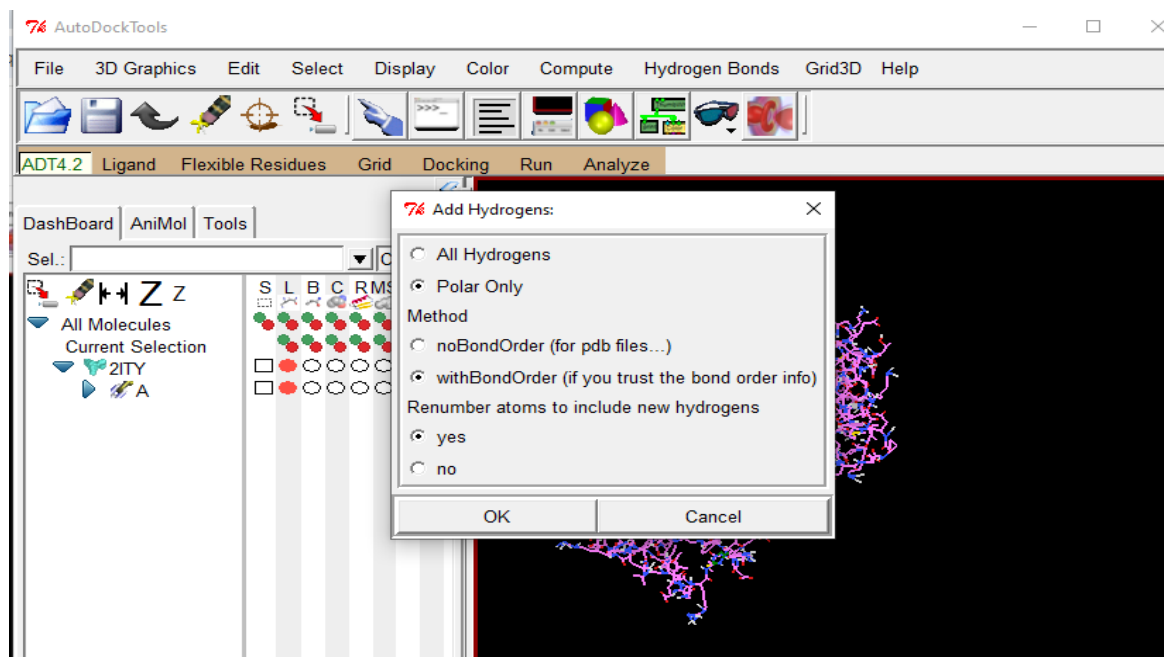


Figure 8: (Polar Hydrogen atoms are added)

2.3 Ligand Preparation

The converted ligand file (Gefitinib.pdb) was loaded into **AutoDock Tools (ADT)** for preparation.

- Launch ADT → Ligand → Input → Open... → Select Gefitinib.pdb.
- This interface allows adding hydrogens, assigning charges, and defining rotatable bonds essential for docking.
- And finally save as .pdbqt.

The following steps will occur-

- **Addition of Polar Hydrogens**- These hydrogens are crucial for hydrogen bonding interactions between ligand and receptor during docking. The nonpolar hydrogens (like those bonded to carbon) are typically ignored to reduce computation time.
- **Assignment of Gasteiger Charges**- After adding hydrogens, **Gasteiger partial charges** were assigned to the ligand atoms to represent the electronic distribution across the molecule. These charges are necessary for accurate **electrostatic energy** calculations during docking. ADT automatically calculates them.
- **Detection of Rotatable Bonds**- The torsional flexibility of the ligand is defined by identifying all **rotatable bonds**. ADT identifies single bonds (except ring bonds) that can rotate. This allows the docking algorithm to explore different ligand conformations for optimal fitting inside the receptor binding site.

Command path:

Ligand → Torsion Tree → Detect Root

then Ligand → Torsion Tree → Detect Rotatable Bonds

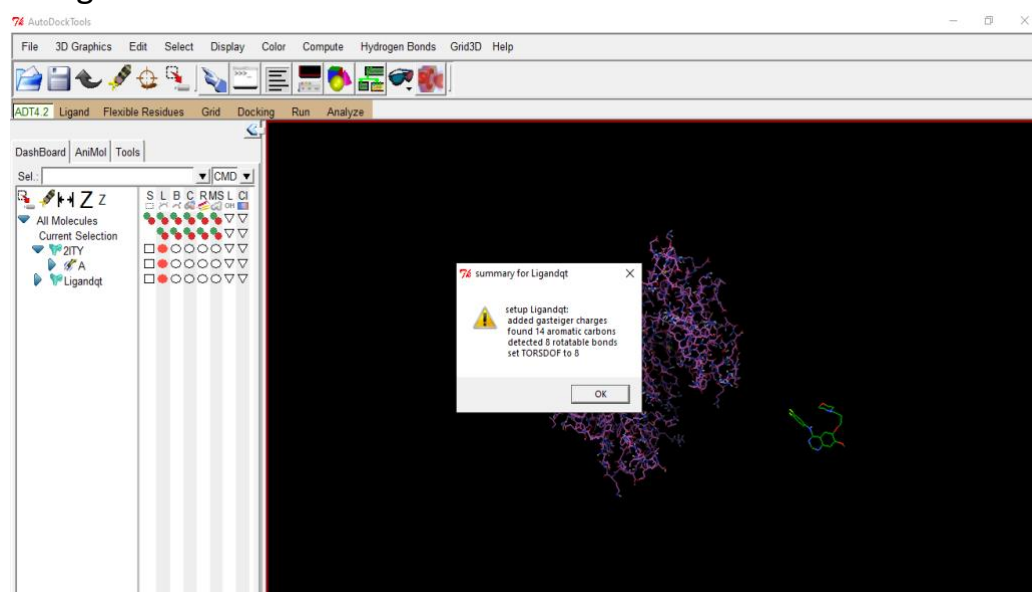


Figure 9: (Ligand Preparation)

2.4 PyRx Molecular Docking

After the receptor and ligand structures were prepared and converted into .pdbqt format using ADT (AutoDock Tools), molecular docking was performed using PyRx (which utilizes the AutoDock engine) to predict the optimal binding conformations and binding affinities of Gefitinib with both wild-type and mutant EGFR receptors.

Step 1: Loading the Prepared Files

The .pdbqt files of the receptor (wild-type and mutant EGFR) and ligand (Gefitinib) were imported into the **PyRx environment**.

Both molecules were visualized in 3D space to verify correct structure and orientation before docking.

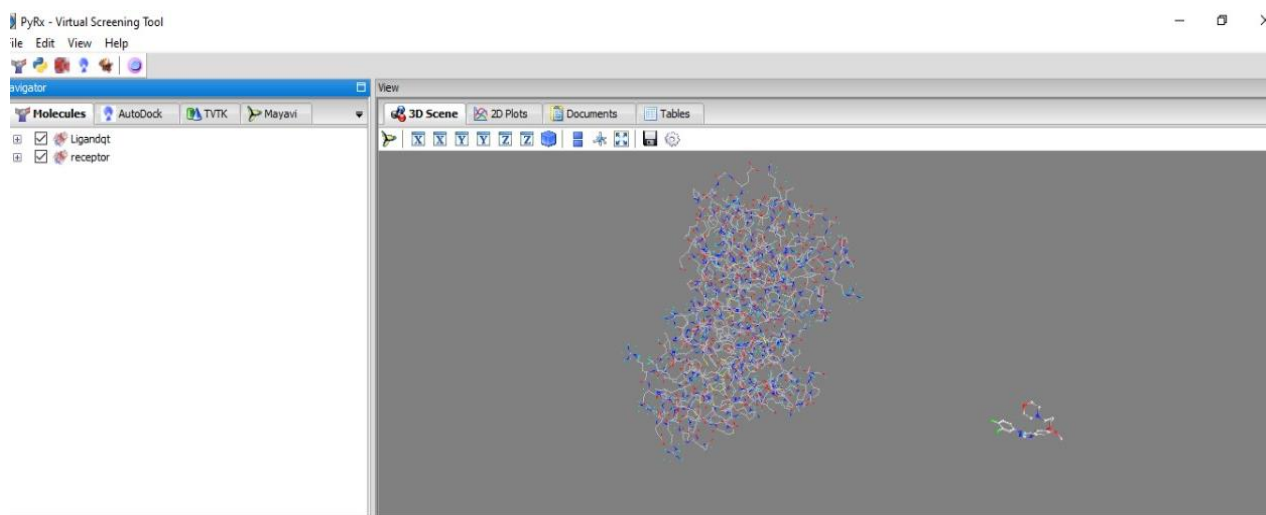


Figure 10: (PyRx with wild-type EGFR and Gefitinib)

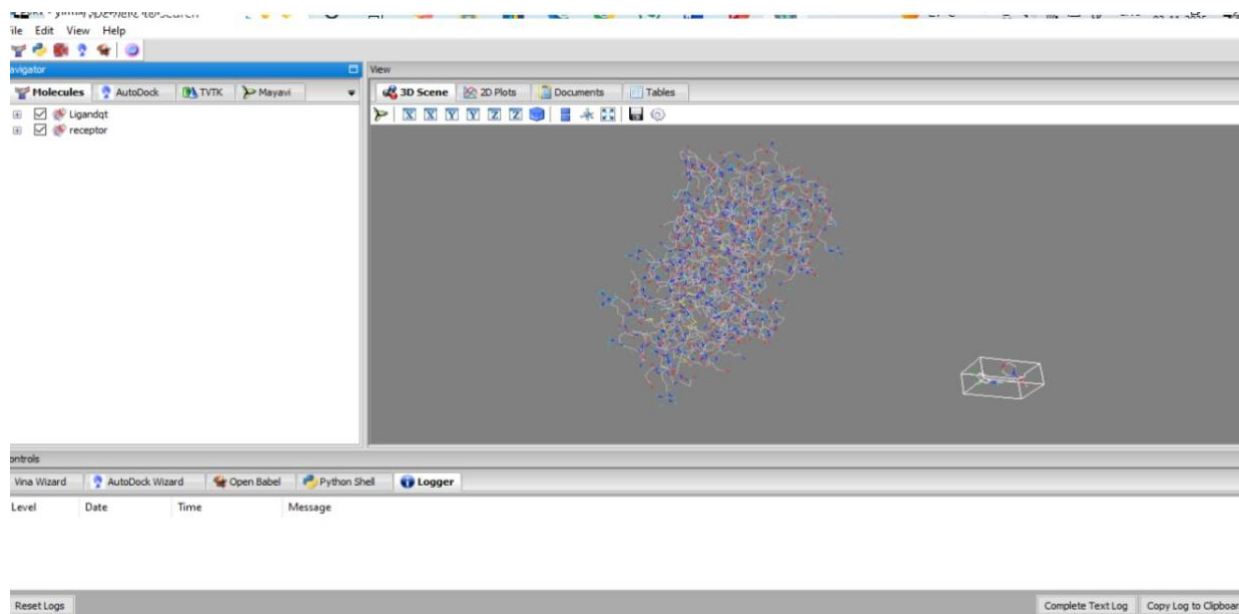


Figure 11: (PyRx with mutant EGFR and Gefitinib)

Step 2: Defining Receptor and Ligand in PyRx

After importing the prepared receptor (EGFR) and ligand (Gefitinib) .pdbqt files into PyRx, both molecules were assigned specific roles for docking.

- The receptor (either wild-type or mutant EGFR) was selected and defined as a macromolecule by navigating to: AutoDock → Make Macromolecule
- The ligand (Gefitinib) was defined as a ligand molecule using: AutoDock → Make Ligand

This classification ensures that AutoDock within PyRx correctly recognizes the receptor as the binding target and the ligand as the binding molecule.

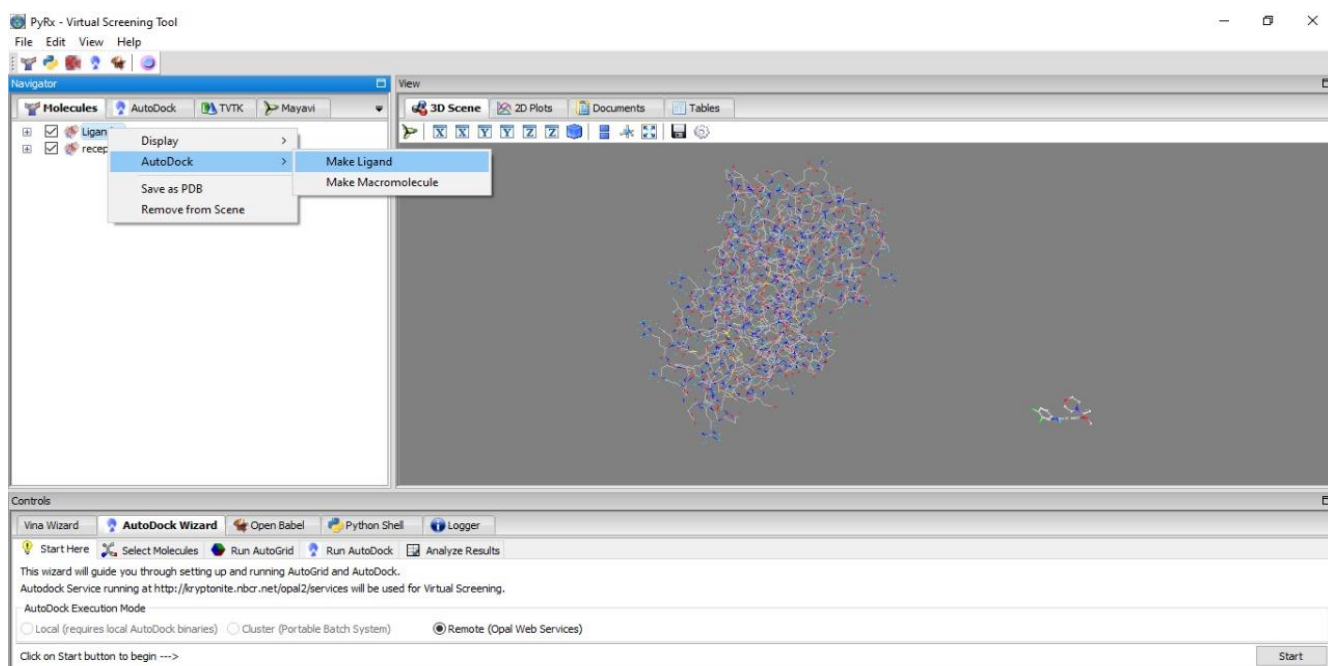


Figure 10: (Classify molecule as Ligand)

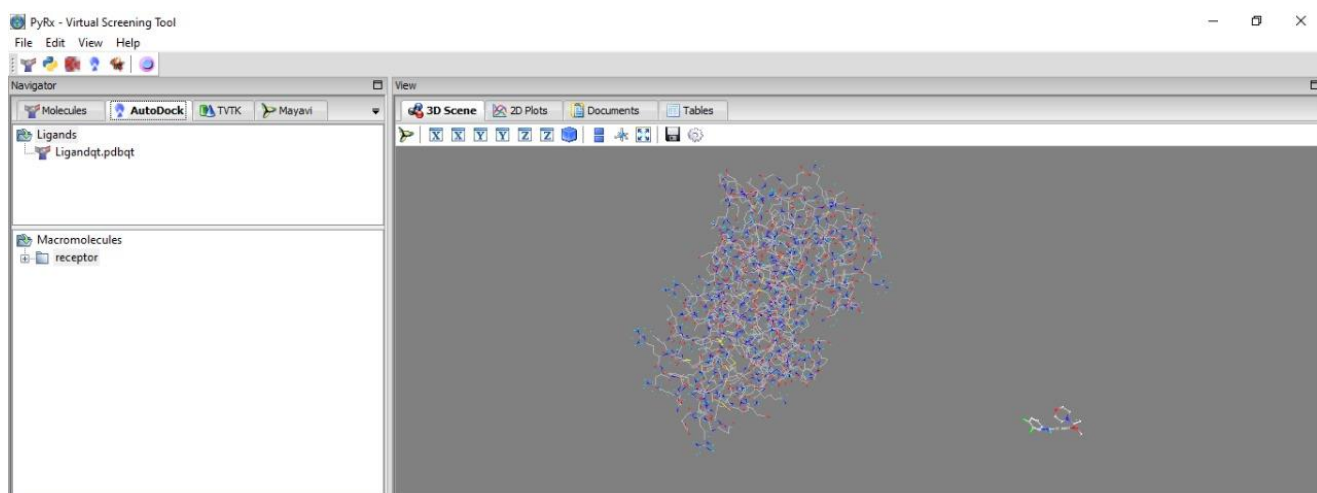


Figure 11: (Both molecules)

Step 3: Grid Box Setup (Active Site Definition)

After assigning the molecules, the AutoDock Wizard was used to define the grid box, which determines the docking search space.

The grid box was adjusted in size and position to enclose the receptor's active/binding site — the region where ligand binding is expected.

And since, in this study blind docking is performed a grid box was defined to encompass the **entire receptor structure** to identify all possible binding sites for the ligand. This approach allows unbiased exploration of ligand–protein interactions without predefining the active site.

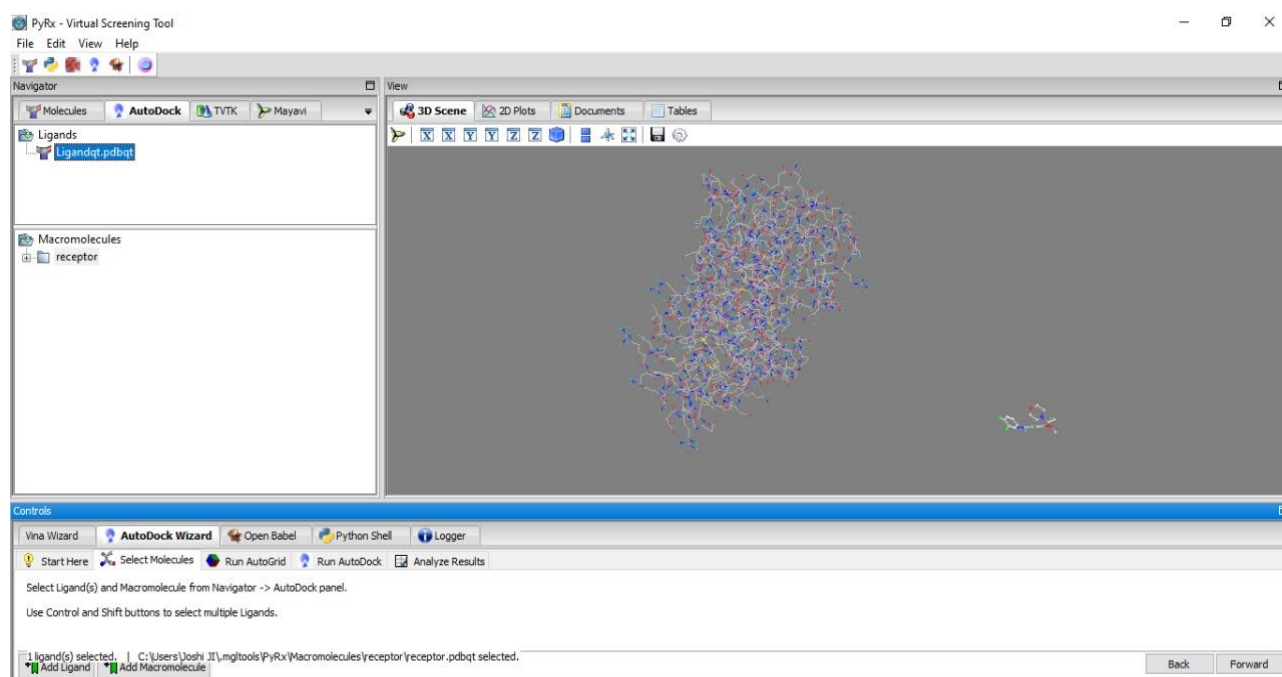


Figure 11: (AutoDock Wizard)

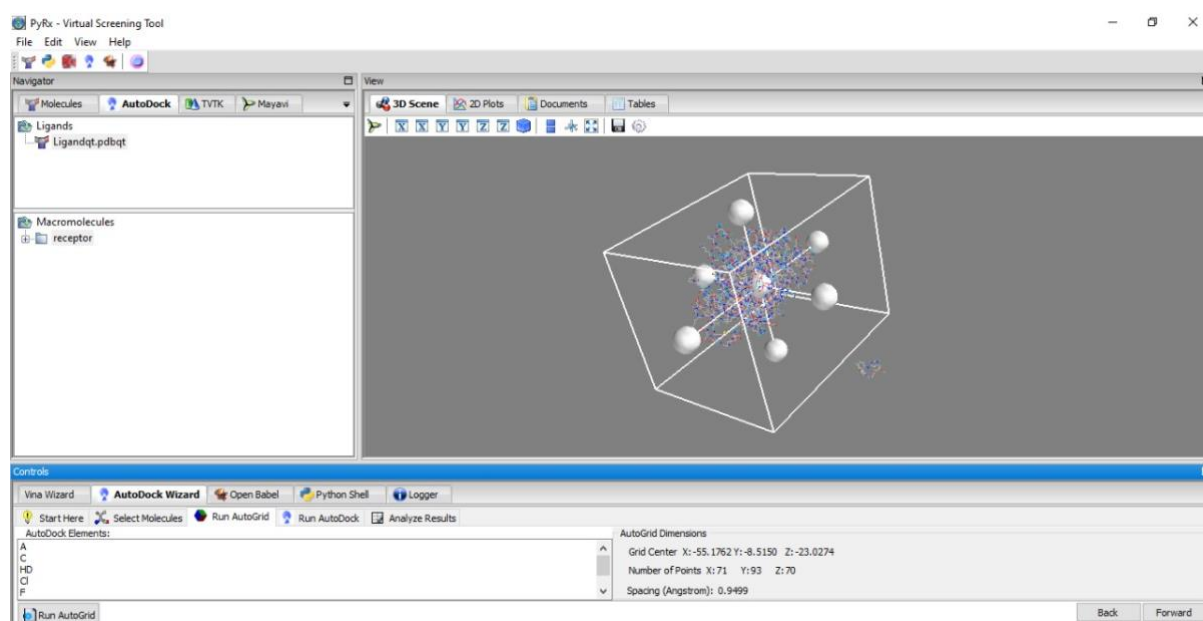


Figure 12: (Grid Box Setup)

Step 4: Running AutoGrid and AutoDock

After confirming the grid box dimensions, the **Run AutoGrid** option was used to pre-calculate potential energy maps for various atom types within the defined grid region. Subsequently, the **Run AutoDock** option was used to initiate the docking process.

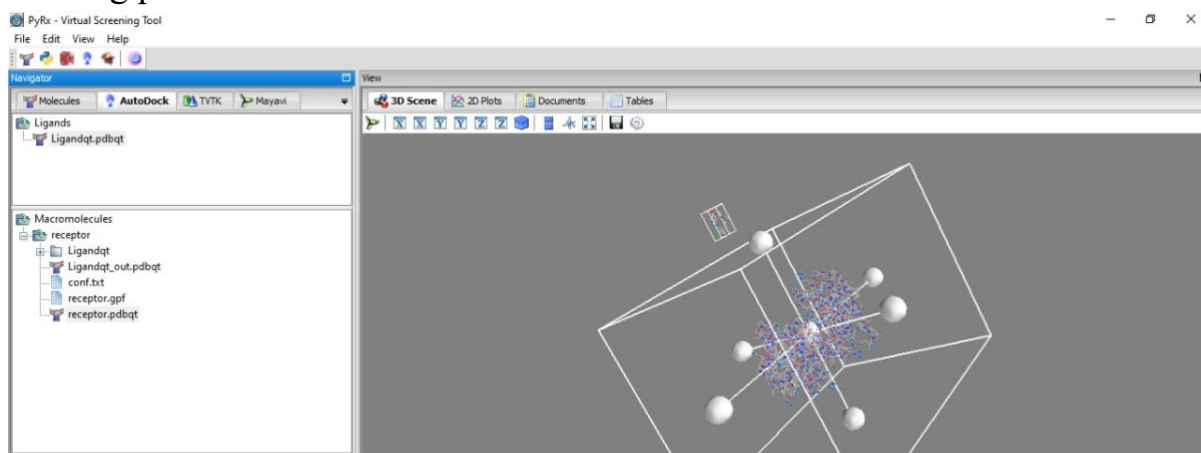


Figure 13: (AutoGrid)

Step 5: Docking Parameters and Algorithm

In the docking setup step within PyRx (**AutoDock Wizard tab**), the **Genetic Algorithm (GA)** was selected as the docking algorithm. This method explores multiple conformations of the ligand within the receptor's binding site by simulating evolutionary principles such as mutation, crossover, and selection. The Genetic Algorithm optimizes the ligand orientation and conformation to achieve the **lowest binding energy pose**, indicating the most stable and favorable interaction between the receptor and the ligand. The default docking parameters of AutoDock were used for this process, and the docking was initiated using the **“Run AutoDock”** option.

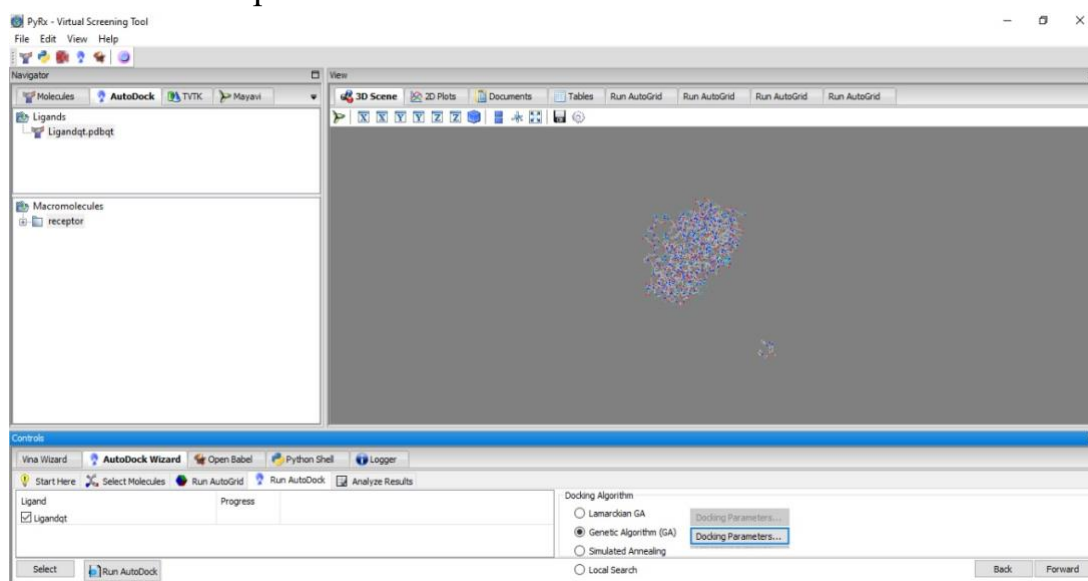


Figure 13: (Docking Parameters Setup)

Step 6: Result Interpretation and Visualization of Docking Poses

The docking results of Gefitinib with EGFR_mt revealed that the best binding pose (Mode 0) exhibited a binding affinity of -6.5 kcal/mol with an RMSD of 0.0 Å, indicating the strongest (>-6 kcal/mol) and most stable interaction. Increasing RMSD values across higher modes represent alternate ligand orientations with slightly weaker binding energies. The top-ranked mode was selected for further visualization in PyMOL to identify key active-site interactions and confirm the ligand's orientation within the EGFR binding pocket.

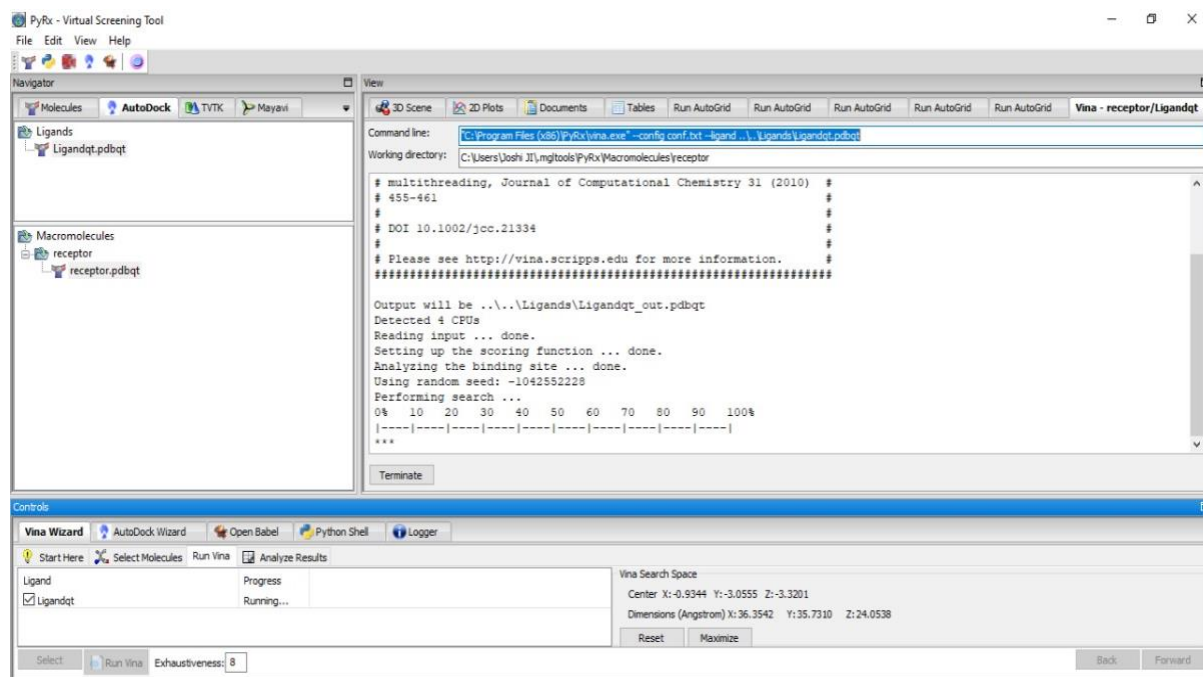


Figure 14: (AutoDock)

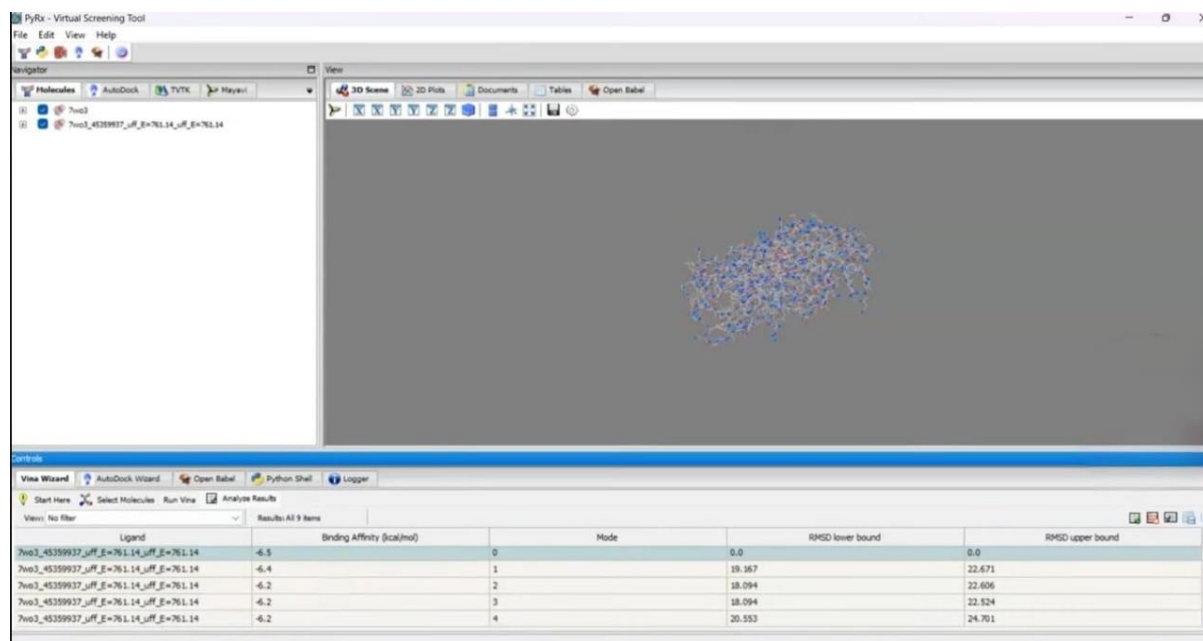


Figure 15 (AutoDock Results)

2.5 PyMOL Molecular Visualization

The best suitable docking pose (Mode 0), having the lowest binding affinity, was loaded into PyMOL for molecular visualization. PyMOL allows detailed examination of the ligand–receptor interactions within the active site of EGFR. Through visualization, the binding orientation of Gefitinib was analyzed to identify key residues involved in the interaction, such as *Met745* and nearby residues around *Thr766*.

Hydrogen bonds, hydrophobic contacts, and other non-covalent interactions were examined to confirm that the ligand is appropriately positioned within the ATP-binding pocket of EGFR. This visualization step helps validate the docking results and provides structural insight into the binding mechanism of Gefitinib.

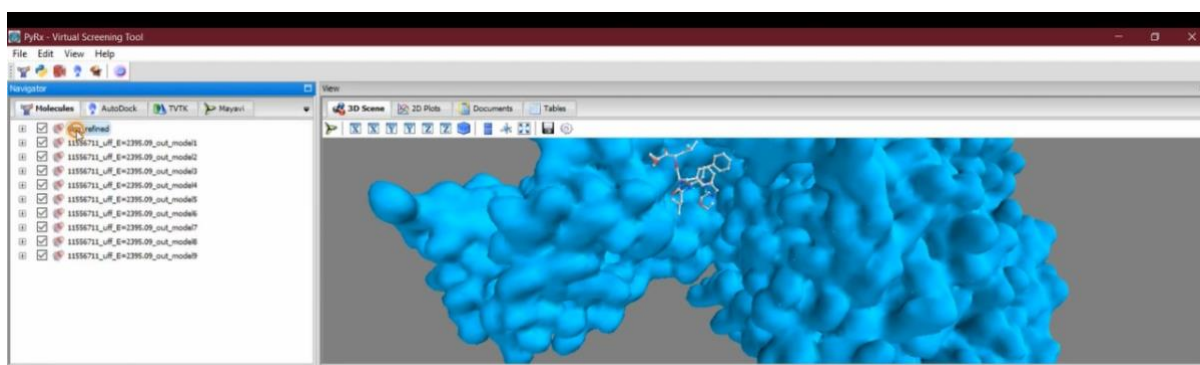


Figure: 16 (Best Mode is selected)

Step: 1 Loading the Best docking pose

The first step in molecular visualization involves loading the best docking pose (Mode 0) of the ligand–protein complex into PyMOL. After performing molecular docking in PyRx (AutoDock Engine), the top-ranked docking conformation showing the lowest binding affinity is exported as a PDBQT or PDB file. This file is then opened in PyMOL for visualization and interaction analysis.

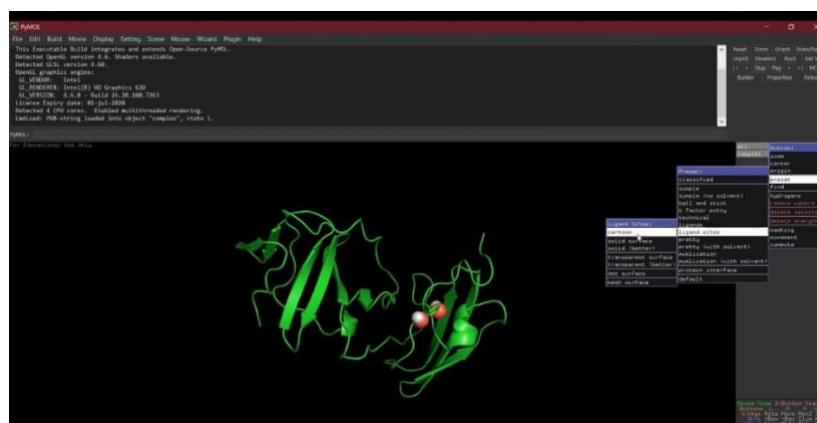


Figure: 17 (PyMOL)

Step 2: Bond Visualization in PyMOL

After successfully loading and orienting the protein–ligand complex, the next step in the molecular visualization process is bond visualization. This step focuses on identifying and analyzing the types of interactions formed between the ligand (Gefitinib) and the active-site residues of the protein (EGFR).

The procedure involves the following detailed steps:

1. Selection of the Ligand and Active Site Residues:

The ligand (Gefitinib) and surrounding residues within approximately 4–5 Å are selected using the *Select* option or the command line

```
> select binding_site, br. (ligand within 5 of protein)
```

This selection focuses the analysis on residues directly involved in binding.

2. Displaying Hydrogen Bonds:

Hydrogen bonds are one of the most significant interactions in ligand binding. In PyMOL, they can be visualized using:

```
> find_pairs (ligand), (protein), mode=1
```

or through the menu “Wizard → Measurement → Find → Polar Contacts”.

Hydrogen bonds are shown as dotted yellow lines, indicating donor–acceptor pairs that stabilize the ligand in the active site.

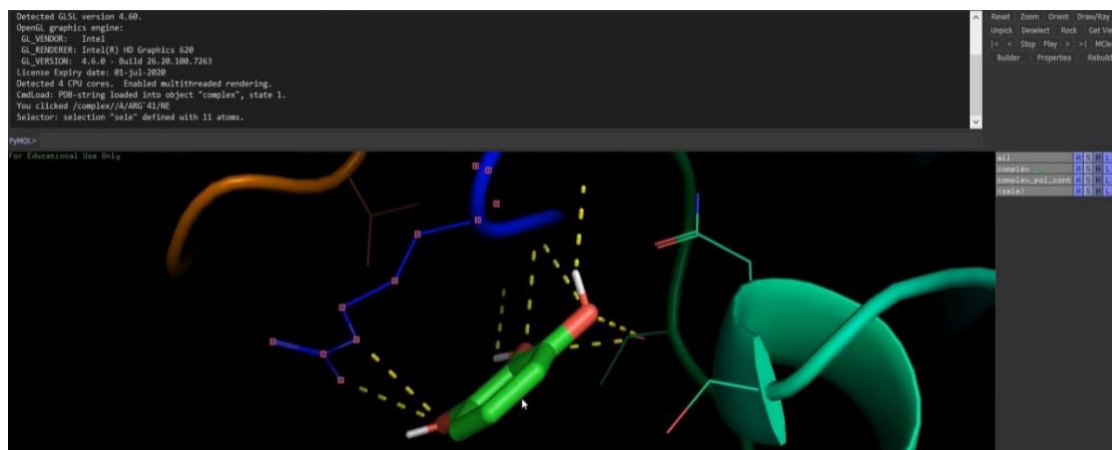


Figure: 17 (hydrogen bonds)

Color Coding for Clarity:

To differentiate interaction types:

- Hydrogen bonds → **yellow dashed lines**
- Hydrophobic residues → **green or grey surface**
- Aromatic/ π -interactions → **magenta sticks**

Step 3: Visualization of Selected Residues in PyMOL

In this step, specific **amino acid residues** from the protein sequence that interact with the ligand (Gefitinib) are **highlighted and displayed** in PyMOL.

After identifying the best docking pose and visualizing the bonds, the **interacting residues** (for example, *Met745*, *Thr766*, *Leu718*, *Val726*, and *Ala743*) can be selected and displayed using the sequence view in the PyMOL.

Procedure:

1. **Opening the Sequence Viewer:**

In PyMOL, the amino acid sequence of the protein can be displayed by selecting “**Display → Sequence**”. This enables viewing of all residues along the protein chain.

2. **Selecting Specific Residues:**

The interacting residues can be selected by clicking directly on them in the **sequence bar**.

3. **Highlighting the Selected Residues:**

Once selected, these residues can be displayed in different representations and colors to emphasize their importance.

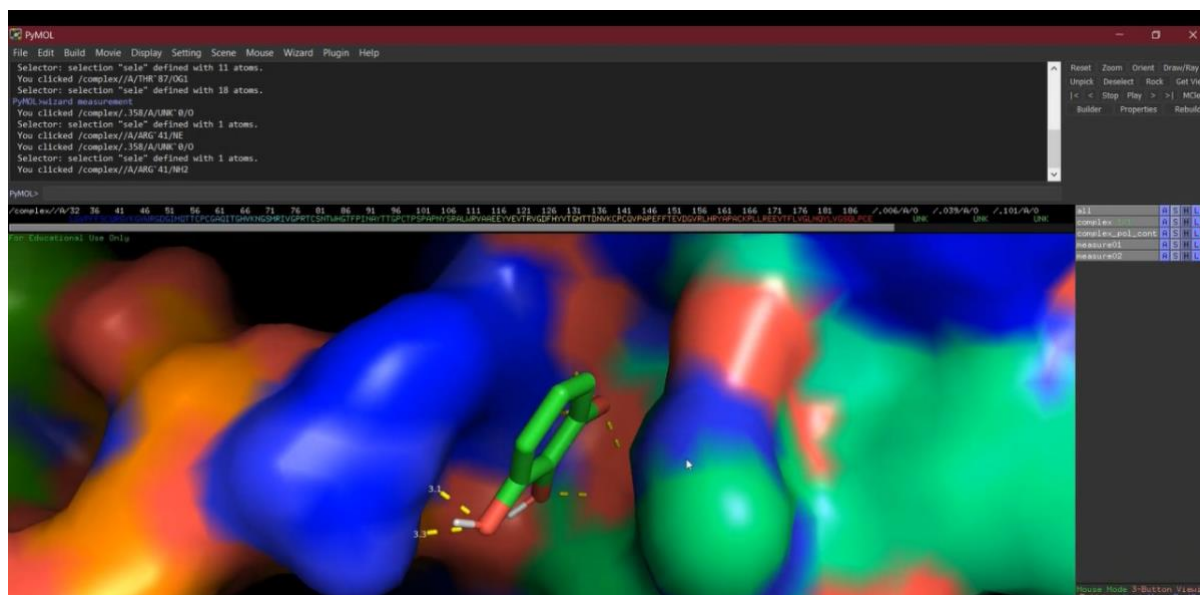


Figure: 18 (molecular surface level format)

Visualization of these residues helps to confirm the location of the ligand within the binding pocket and to understand the specific residue-level interactions.

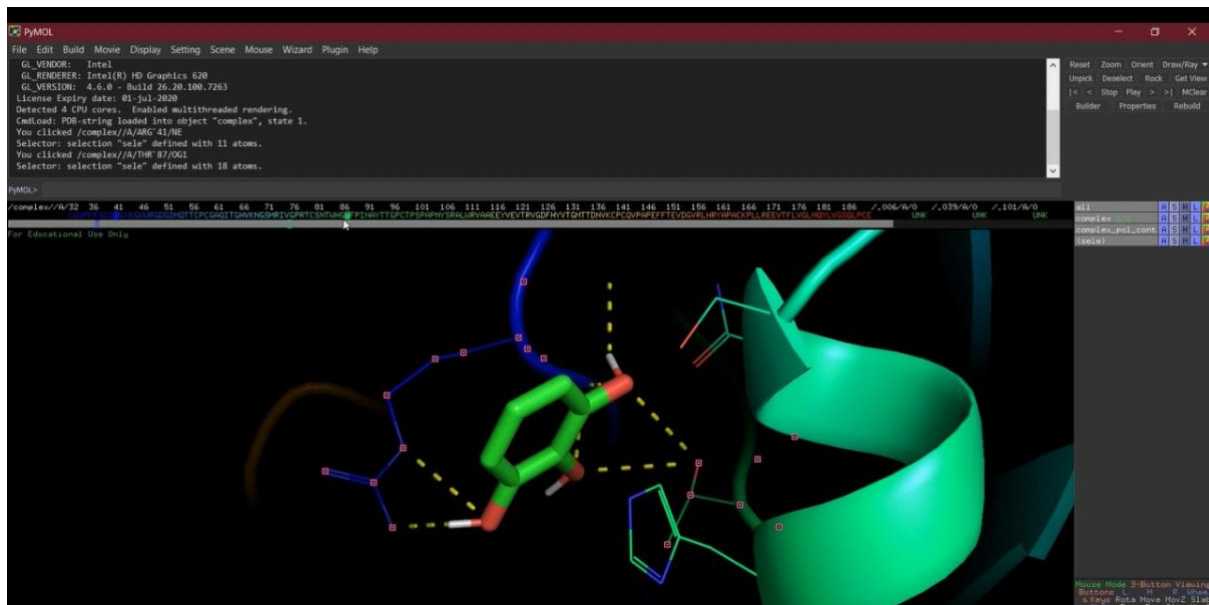


Figure: 19 (cartoon format)

CHAPTER- 3 ANALYSIS AND INTERPRETATION

3.1 Results of Molecular Docking (PyRx)

The pose with the lowest (most negative) binding affinity represents the most stable conformation and strongest predicted interaction between the ligand and the target protein. The top-ranked pose displayed a binding affinity of -6.5 kcal/mol, indicating a favorable interaction. The RMSD (Root Mean Square Deviation) values between modes were also examined; a lower RMSD (close to 0 Å) between poses suggests consistent and reliable docking results. Higher RMSD values indicate alternative binding orientations within the binding site.

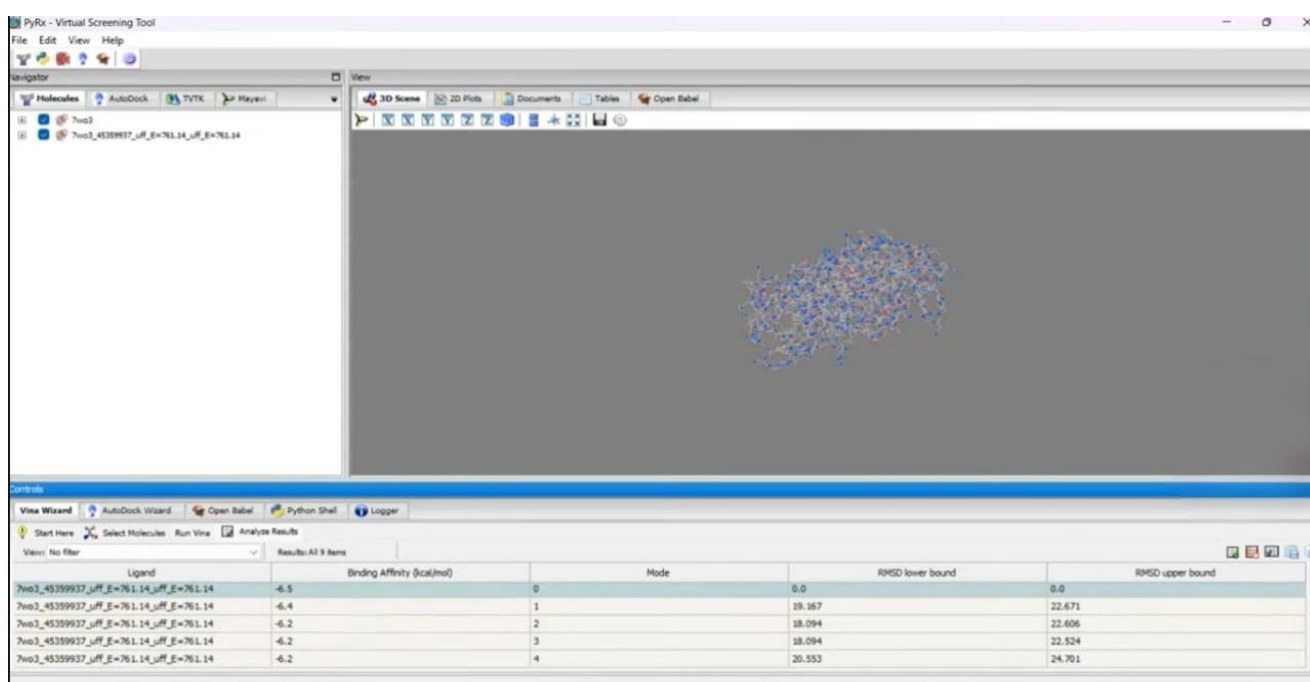


Figure: 20 (AutoDock Results 1)

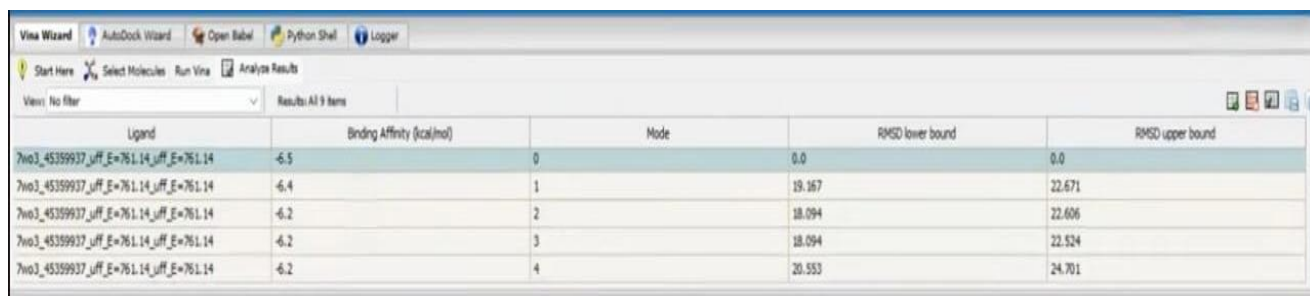


Figure: 21 (AutoDock Results 2)

3.1.2 Result Table Interpretations

Column	Meaning
Ligand / Complex	The ligand (Gefitinib) docked with the protein (EGFR wild-type).
Binding affinity (kcal·mol ⁻¹)	The predicted free energy of binding — more negative values indicate stronger binding and greater stability of the ligand–receptor complex.
Mode	Each docking mode represents a possible binding conformation (pose) of the ligand in the active site. Mode 0 is the best (top-ranked) conformation with the lowest binding energy.
RMSD lb (Å)	RMSD = <i>Root Mean Square Deviation (lower bound)</i> — measures the difference between the heavy-atom positions of the current pose and the best pose (mode 0). A small RMSD (< 2 Å) means the pose is similar to the best one.
RMSD ub (Å)	RMSD = <i>Root Mean Square Deviation (upper bound)</i> — indicates the maximum atomic displacement compared to the reference pose. Larger RMSD values suggest different ligand orientations.

Table 1: PyRX Result Interpretations

The **most stable binding conformation** is **Mode 0** with a **binding affinity** of –6.5 kcal/mol, indicating the **strongest predicted interaction** between Gefitinib and the EGFR receptor. Other modes (1–8) have slightly less negative binding energies (–6.5 to –5.3 kcal/mol) and increasing RMSD values, meaning they are **alternative poses** or orientations of the ligand within the binding pocket.

The **best docking pose (Mode 0)** — having the **lowest binding affinity** — is then **selected for visualization in PyMOL** to study:

- The **position of Gefitinib** within the EGFR active site.
- The **key interacting residues**, such as *Met745* and residues around *Thr766*, that stabilize the ligand through hydrogen bonds or hydrophobic interactions.

This visualization helps confirm whether Gefitinib binds correctly at the **ATP-binding site** of EGFR and forms interactions consistent with experimental structures.

CHAPTER- 4 FINDINGS AND SUGGESTIONS

4.1 Key Finds of Project

The Epidermal Growth Factor Receptor (EGFR) is a receptor tyrosine kinase that becomes activated upon binding extracellular ligands such as EGF, TGF- α , and amphiregulin, leading to dimerization and kinase activation. Within the intracellular kinase domain, ATP binds to the catalytic pocket and donates phosphate groups for trans-autophosphorylation, triggering downstream signaling cascades.

Mutations commonly associated with non-small cell lung cancer (NSCLC) — including exon 19 deletions and the L858R point mutation — occur within the kinase domain. These mutations result in ligand-independent constitutive activation of EGFR, which sustains downstream RAS–MAPK and PI3K–AKT signaling pathways, promoting uncontrolled cell proliferation.

Gefitinib, an ATP-competitive tyrosine kinase inhibitor (TKI), targets the ATP-binding site of EGFR. By occupying this site, it prevents ATP from binding, thereby blocking autophosphorylation and downstream signaling in Gefitinib-sensitive mutant receptors.

Receptor Type	PDB ID	Binding Affinity (kcal/mol)	Key Interacting Residues	Type of Interactions	Interpretation
Wild-Type EGFR	1M17	-5.8	Met793, Leu718, Ala743, Lys745	Hydrogen bonding, hydrophobic contacts	Moderate binding; baseline interaction pattern within ATP-binding pocket
Mutant EGFR (L858R)	2ITY	-6.5	Met793, Leu858, Asp855, Thr790, Lys745	Strong hydrogen bonds, π - π stacking, hydrophobic interactions	Enhanced binding affinity; mutation increases pocket flexibility and stabilizes drug binding

Table 2: Findings

Comparative molecular docking and visualization of the wild-type and mutant EGFR–Gefitinib complexes provided mechanistic insight into drug binding affinity, selectivity, and mutation-induced conformational changes. The results revealed that Gefitinib binds more strongly and stably to the mutant EGFR compared to the wild-type receptor, suggesting that structural alterations in the mutant kinase domain enhance the inhibitor’s affinity. These computational findings emphasize the distinct differences in binding affinity and interaction patterns between the two receptor types. Stronger binding of Gefitinib to the mutant receptor reflects its therapeutic selectivity toward NSCLC-associated EGFR variants. The study also highlights key stabilizing interactions at the mutation site, which likely contribute to the enhanced binding stability observed.

4.2 Suggestions

From a clinical perspective, such computational analyses could complement molecular diagnostic methods by predicting mutation-induced changes in EGFR binding behavior. Enhanced Gefitinib binding to mutant EGFR may serve as a molecular marker for oncogenic activation and potential responsiveness to TKI therapy. Overall, the study establishes a computational framework for understanding mutation-driven variations in EGFR structure, drug affinity, and stability, offering valuable insights for structure-based drug design and personalized cancer therapeutics.

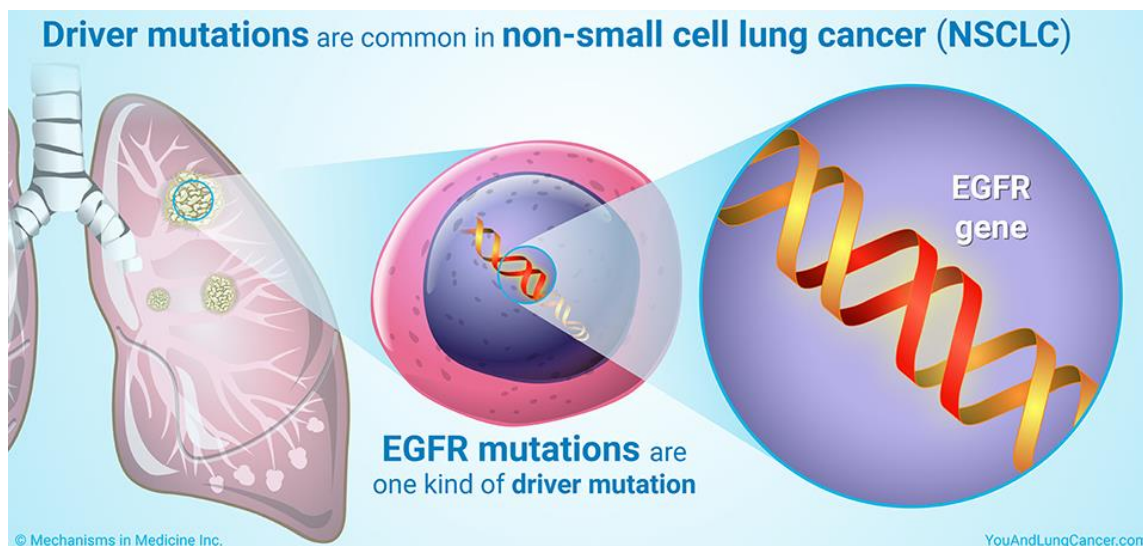


Figure: 22 (Lung Cancer) [6]

CHAPTER-5 CONCLUSION

The comparative computational analysis between the wild-type and mutant EGFR–Gefitinib complexes revealed distinct differences in binding affinity and interaction patterns. The mutant EGFR exhibited stronger binding and higher stability, reflecting its enhanced susceptibility to Gefitinib inhibition. These results confirm that EGFR mutations significantly influence the drug–receptor interaction dynamics and support Gefitinib’s therapeutic selectivity toward NSCLC-associated mutants.

This study provides a molecular-level understanding of the structural basis of EGFR–Gefitinib interaction and demonstrates how mutation-induced conformational changes affect drug binding.

Furthermore, in the clinical context, these insights could assist in precision oncology, enabling the design of mutation-specific inhibitors and supporting personalized therapeutic strategies for lung cancer patients.

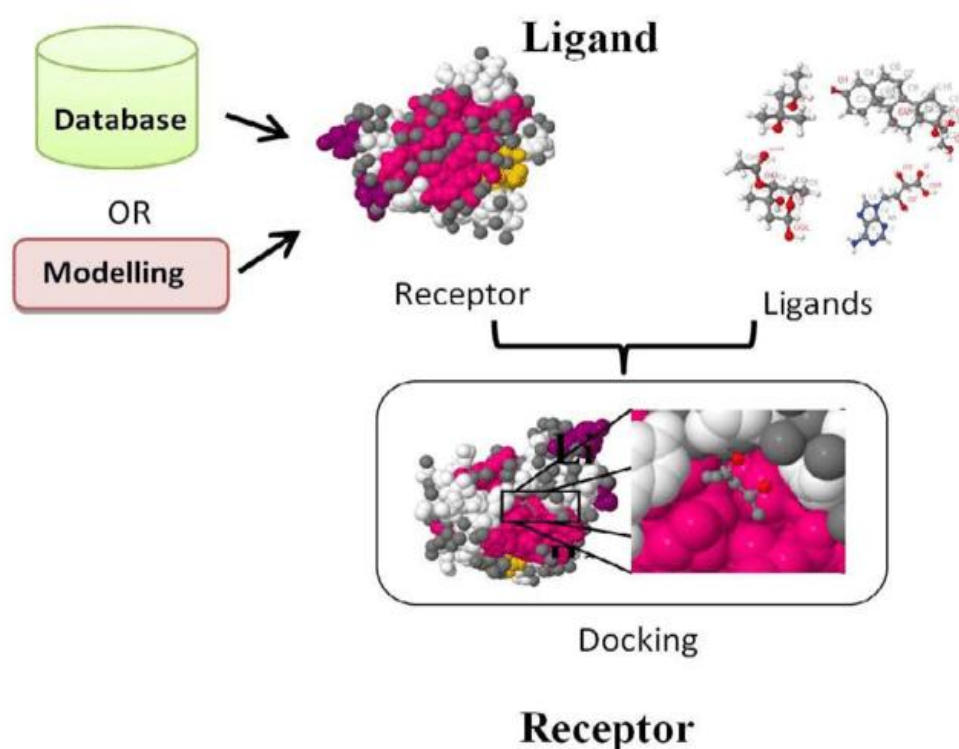


Figure: 23 (Molecular Docking) [7]

CHAPTER- 6 REFERENCES

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CHAPTER- 7 APPENDIX

7.1 Appendix: Software and Tools Used

Tool/Database	Purpose	Source/URL
PyRx (v0.8)	Virtual screening and molecular docking (uses AutoDock as backend)	https://pyrx.sourceforge.io
AutoDock Vina	Docking engine employing Genetic Algorithm (GA) for ligand-receptor binding	https://vina.scripps.edu
PyMOL	Molecular visualization and 3D structural analysis of docked complexes	https://pymol.org
Protein Data Bank (PDB)	Source of EGFR receptor structures (wild-type and mutant)	https://www.rcsb.org
PubChem	Source of Gefitinib ligand structure (CID: 123631)	https://pubchem.ncbi.nlm.nih.gov

Table 3: Appendix